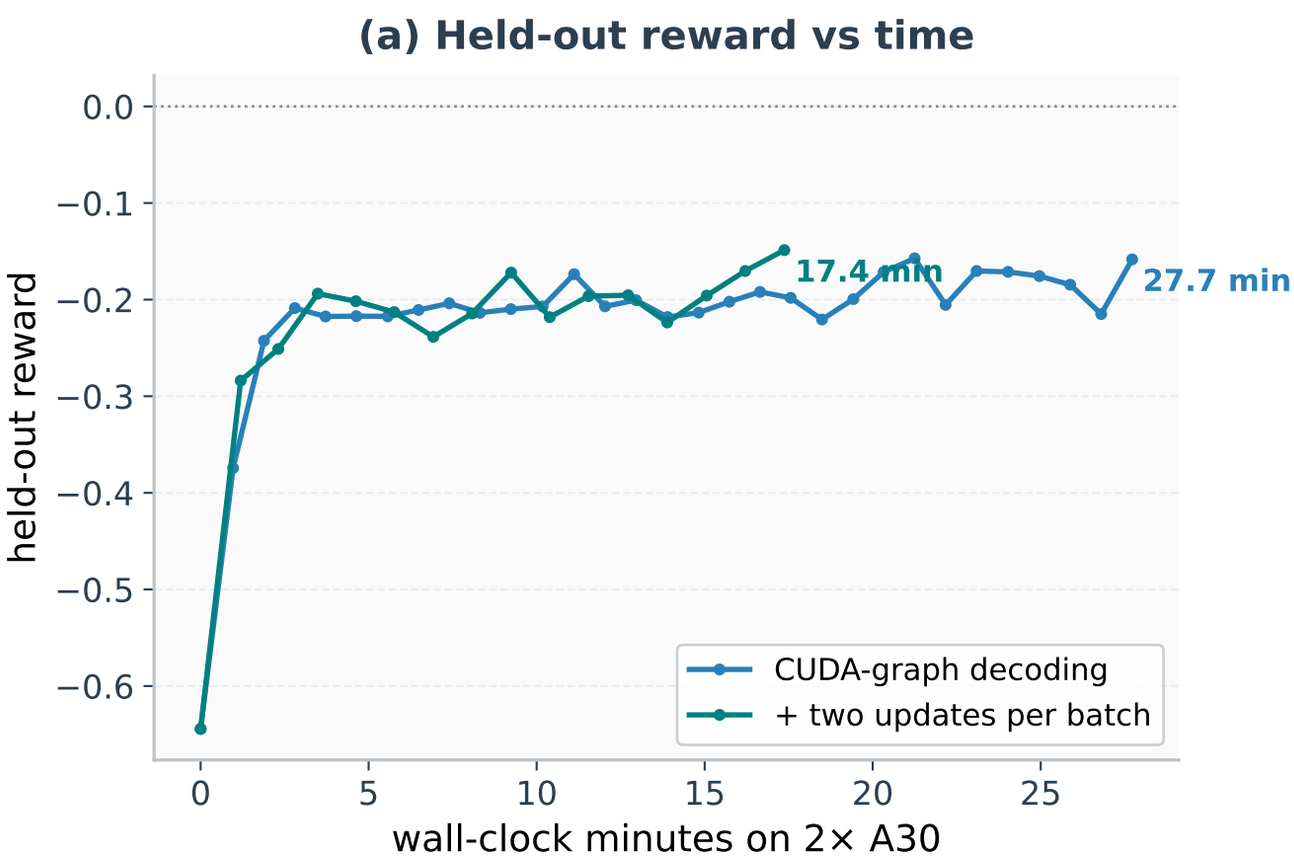
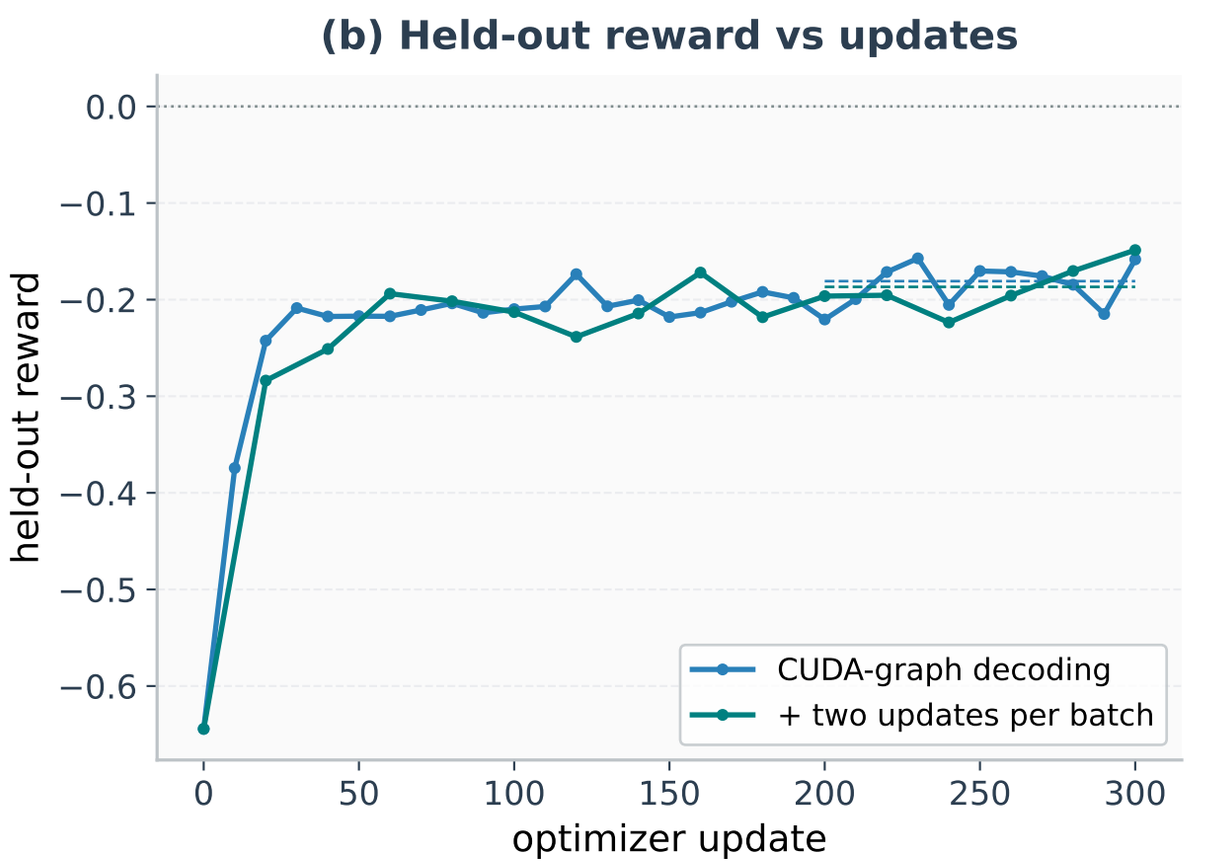
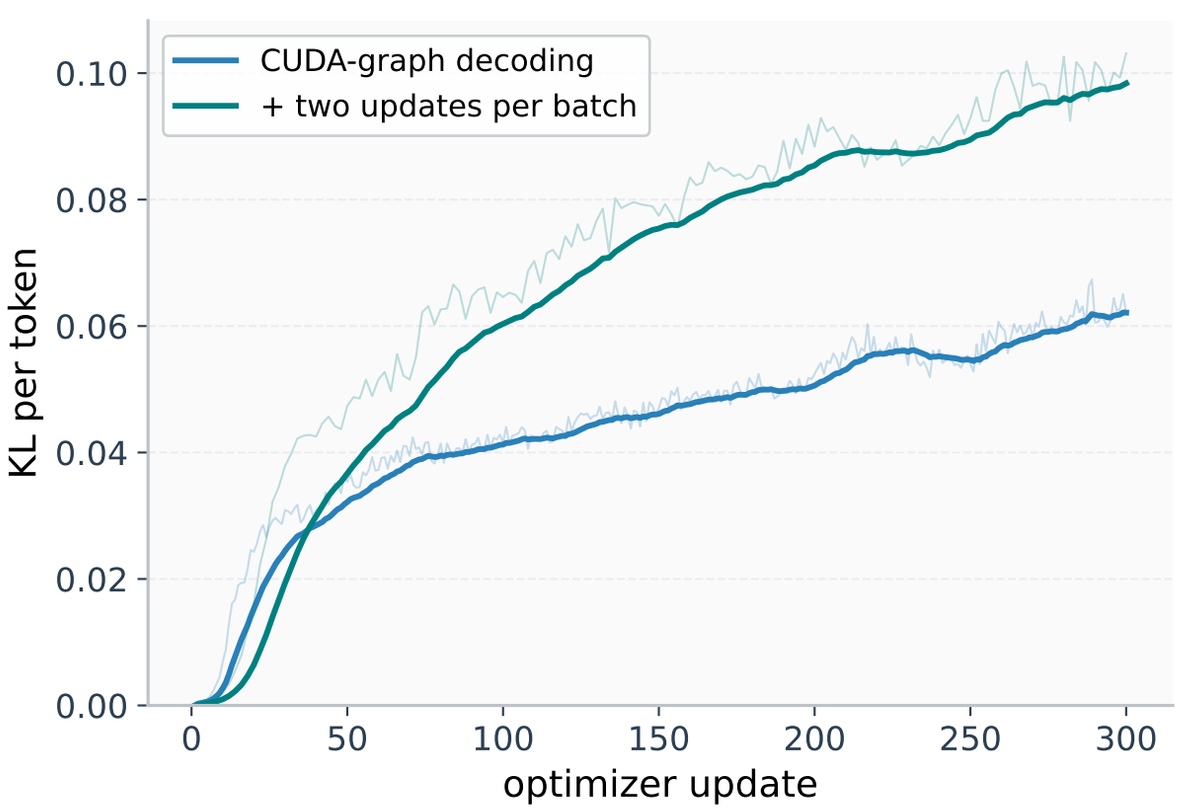
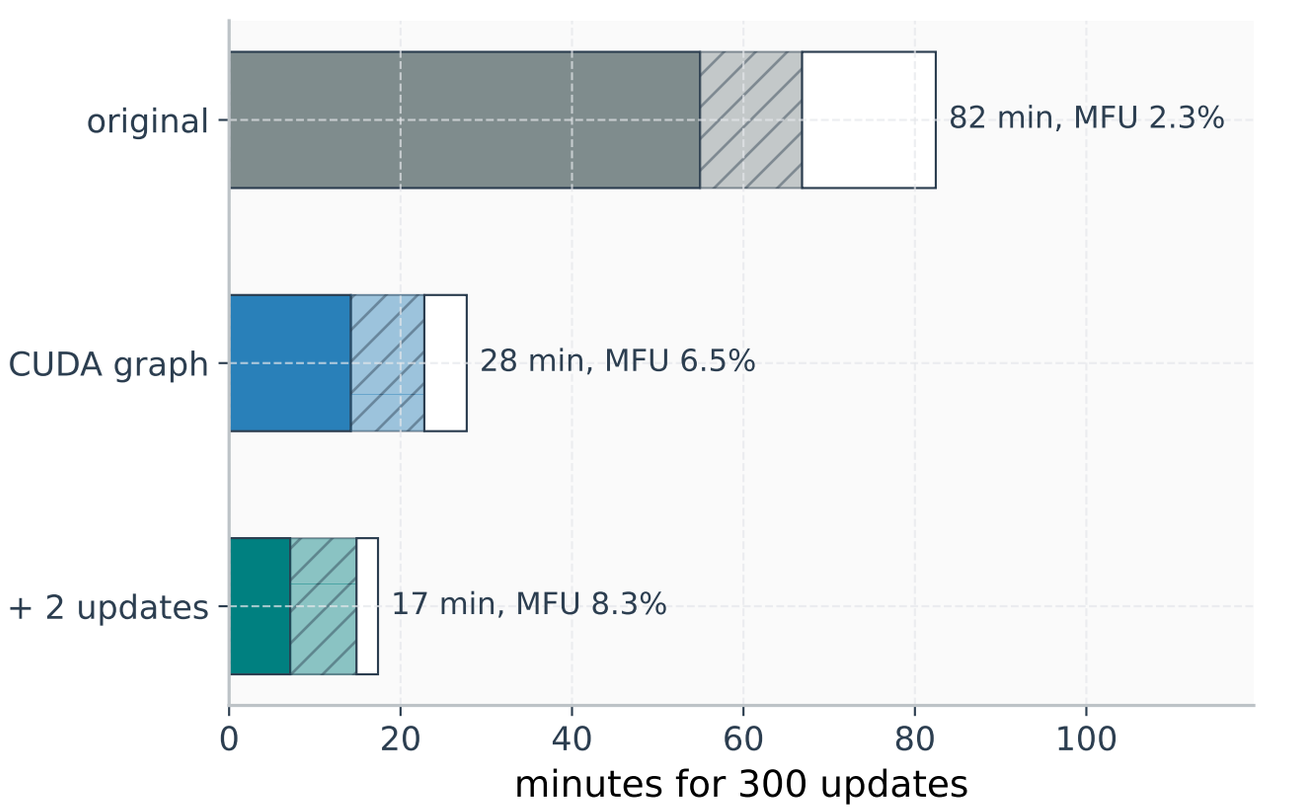

(a) Held-out reward vs time
held-out reward
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
17.4 min
27.7 min
CUDA-graph decoding
+ two updates per batch
0
5
10
15
20
25
wall-clock minutes on 2× A30


(b) Held-out reward vs updates
held-out reward
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
CUDA-graph decoding
+ two updates per batch
0
50
100
150
200
250
300
optimizer update


(c) KL from the SFT model
CUDA-graph decoding
+ two updates per batch
KL per token
0.10
0.08
0.06
0.04
0.02
0.00
0
50
100
150
200
250
300
optimizer update


(d) Where the time went
original
CUDA graph
+ 2 updates
82 min, MFU 2.3%
28 min, MFU 6.5%
17 min, MFU 8.3%
0
20
40
60
80
100
minutes for 300 updates
generation
policy update
evals and checkpoints